

Soil Sidekick Pro

Client Quality Report

Report Date: February 26, 2026

1) Scope (What This Report Covers)

- **Code snapshot:** `main` @ commit `74a6288`
 - **Test window:** February 26, 2026
 - **Environments checked:**
 - Local automated frontend/unit test run
 - Production API spot checks for 2 backend services
 - **In scope:**
 - Automated test suite health
 - Coverage gap status
 - Validation/performance spot checks for `county-lookup` and `carbon-credit-calculator`
 - **Out of scope:**
 - Full load/performance testing under sustained traffic
 - Full security penetration testing
 - End-to-end mobile device testing
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2) Executive Summary (Measured)

Metric	Value
Automated tests executed	139
Passing	138
Failing	1
Pass rate	99.3%
Backend function directories	43
Backend function test files	2 (4.7% coverage)
Page-level frontend files	47
Page-level test files	0

Bottom line: Core tested areas are stable, but backend test coverage remains the largest quality gap.

3) Trend vs Previous Report

Metric	Previous (Feb 10, 2026)	Current (Feb 26, 2026)	Delta
Frontend tests counted	111	139	+28
Frontend pass/fail	111/111	138/139	+27 pass, +1 fail
Estimated line coverage	~22%	~22% (no new measured run)	No confirmed change
Edge-function test files	2	2	0
Edge functions reported	42	43 directories	+1

4) What I Validated This Cycle

4.1 Test Suite Health

- Command run: `npx vitest run`
- Result: **13 passed files, 1 failed file**
- Failing case:
 - `src/hooks/__tests__/useCostMonitoring.test.tsx`
 - Cause: Expected old error-toast copy; app now returns updated copy.

4.2 Backend Service Checks

county-lookup (Production Spot Checks)

Test	Result
Valid request	200
Invalid short input	400
CORS preflight	200

Performance sample (sequential warm checks, n=5):

- Server response-time avg: **1723 ms**
- P95: **2285 ms**
- Range: 1153–2285 ms
- Target: **<1000 ms** (current sample exceeds target)

carbon-credit-calculator (Production Spot Checks)

Test	Result
Missing auth header	401
Anon bearer token request	401
CORS preflight	200

Interpretation:

- Access control for unauthorized requests is functioning.
 - Current tests focus mainly on error responses; authenticated business-path coverage is still missing.
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5) Security Statement

- Unauthorized request without auth header blocked (**401**) – `carbon-credit-calculator`
- Request with anon bearer token blocked (**401**) – `carbon-credit-calculator`
- Invalid short-term query rejected (**400**) – `county-lookup`
- CORS preflight handling successful (**200**) for both services

Not covered this cycle:

- Expired JWT scenarios
 - Role escalation attempts
 - Cross-tenant data-access attempts
 - Replay attack simulation
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6) Product-Facing Risk Scenarios

1. Backend coverage gap

- Risk: Regression in untested high-use endpoints (e.g., `get-soil-data`) reaches production.
- Impact: Intermittent failures or incorrect results in key workflows.

2. County lookup latency risk

- Risk: Response times >1.5s under user traffic.
- Impact: Slower search experience and reduced task completion.

3. Reporting consistency risk

- Risk: Stakeholders reference outdated test counts.
 - Impact: Misalignment in readiness decisions.
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7) Decisions Requested

1. County lookup optimization now?

- **Benefit:** Faster search experience
- **Tradeoff:** May delay one planned backend test suite by 2–3 days

2. Backend test priority approval (**get-soil-data** → **trial-auth** → **api-key-request**)?

- **Benefit:** Reduces highest production risk first
- **Tradeoff:** Lower-priority endpoints remain untested longer

3. Documentation baseline reset to current metrics?

- **Benefit:** Avoids stakeholder confusion
 - **Tradeoff:** Short one-time cleanup effort
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Appendix A: Internal Team Questions (Prioritized)

A1) Urgent

1. Which latency threshold should gate `county-lookup` in CI (hard fail vs warning-only)?
2. Do we have a stable professional-tier test account for authenticated `carbon-credit-calculator` success-path tests?
3. Should updated toast copy be locked as the product standard and tests updated accordingly?

A2) Important

1. Should `sandbox-demo` checks be moved to a separate suite for cleaner ownership?
2. Should edge-function coverage be tracked by function directory count or deployable endpoint count?

A3) Nice to Have

1. Should a monthly quality trend dashboard (tests, failures, latency, coverage) be published for client reviews?